

Infix to prefix algorithm

27 April 2023 11:58

- 1) start scanning from **right to left**
- 2) If(current element is an operand)
Append it to the prefix expression

Else

// If(current element is an operator)

While(stack is not empty && priority of topmost element > priority of current element)

```
{
    Pop element from the stack and append it to prefix
    expression
}
```

Push current element onto the stack.

Infix expression:

- $5*3/2*6/4+7-3+8/2$

Stack :

Current element :

Prefix Expression: 28/3746235*/*/+--

Reverse: +--/*/*5326473/82

- 3) Repeat the above steps till end of infix expression
- 4) Pop all the remaining elements from the stack one by one and append them to prefix expression
- 5) **Reverse the prefix expression**

()
Order of
*/
+-

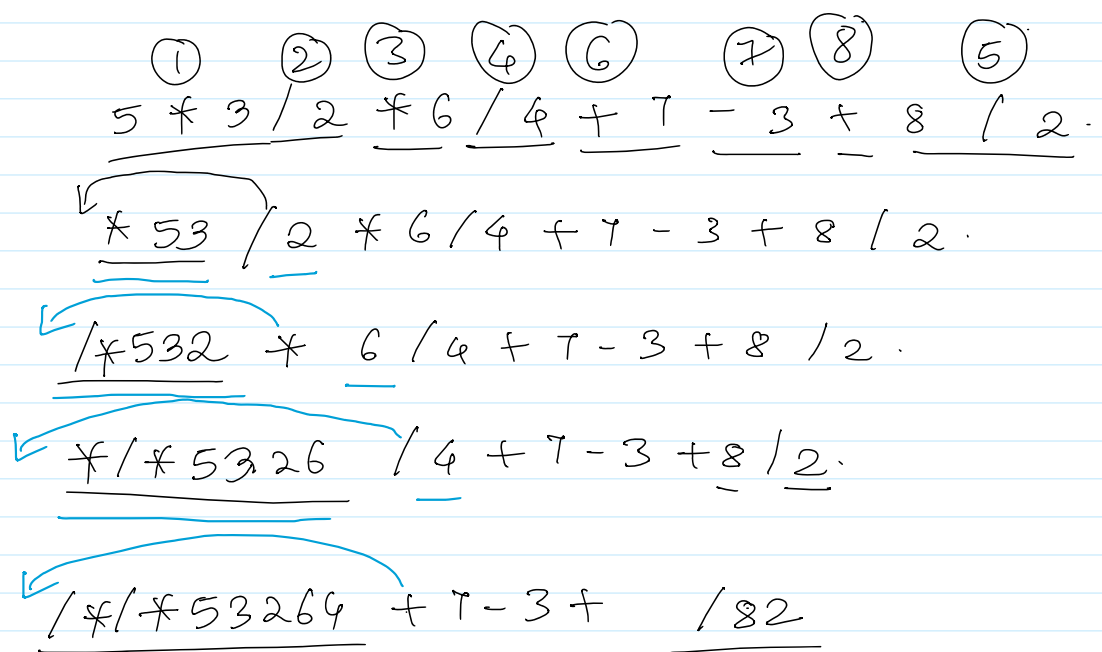
If the current element is the closing bracket,

Push it on to the stack

If the current element is the opening bracket,

- Pop elements from the stack and append them to prefix exp till its corresponding closing brackets does not occur
- Pop opening brackets from the stack and discard both the brackets

- $5*3/2*6/4+7-3+8/2$



$$\frac{(*/*53264 + 7 - 3 + /82}{}$$

$$\frac{+/*/*532647 - 3 + /82}{}$$

$$\frac{- +/*/*5326473 + /82}{}$$

$$\boxed{+-+/*/*5326473/82.} \quad \text{prefix.}$$

$$+-+/*/*5326473/82$$